

torch-spyre Ecosystem Health Report

Date: 2026-05-28 | Pod: rganti-spyre-main | PyTorch: 2.11.0+cpu

3/3 BUILD PASS	PASS NIGHTLY CI	30+ PRS MERGED (7D)	30 OPEN BUGS
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What Needs Attention

- 1

PR #2218 (Upgrade to PT 2.12) still idle — Open since 2026-05-21, REVIEW_REQUIRED. Cross-repo PyTorch version bump deserves reviewer attention given upstream impact.
- 2

Quick wins stalled in review — #1440 (cpu_compile flags) and #1444 (linters.yaml typo) are APPROVED but unmerged since 2026-04-08. ~30 PRs are >7 days stale.
- 3

Two new infra bugs from mudhakar — #2246 (propagate_hints thread safety) and #2247 (positional matching) target hint plumbing; thread-safety implications warrant near-term attention.
- 4

Confirm nightly stability tomorrow — Today's nightly recovered after 4 consecutive failures (cat zero-tensor regression). Watch tomorrow's run to rule out flakiness.
- 5

Heavy upstream churn in deeptools (82 commits, 7d) — Build clean today, but BaseOpt/RcuOpt allocator and DCC ProgIR changes warrant continued monitoring; flex FlexAllocator API consolidation is in flight.

Build Status

Component	Commit	Status	Notes
deeptools	d13b64bf5a — [Util] revert -1M saturation in FloatToFp16BinInternal()	PASS	Incremental cmake + make -j16 install (~1m 11s)
flex	a233ca5 — Add region_id to segment_id mapping to FlexAllocator (#1034)	PASS	Cached configure + install (~12s)

torch-spyre	57da23f — Fix core-div-mismatch logic with PerCoreView (#2298)	PASS	uv sync editable install (~2m 18s). Total stack build wall time: ~3m 41s
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CI Status

Workflow	Latest	Status	Trend
Spyre Hardware Tests	2026-05-28 11:32 UTC	PASS	Healthy (success, success, cancelled)
Linters	2026-05-28 11:32 UTC	PASS	Healthy (one isolated failure mid-cycle, recovered)
Upstream PyTorch Tests	2026-05-28 11:32 UTC	PASS	Healthy
Nightly Tests	2026-05-28 03:16 UTC	PASS	Recently recovered after 4-day cat zero-tensor failure streak

Nightly Failure Triage

Status: PASSING (recovered today after 4 consecutive failures)

Today's run (2026-05-28): 2293 passed, 54 skipped, 2 previously-failing tests now green.

Resolved tests: test_cat_4d_dim2_zero, test_cat_4d_dim_m2_empty_first (in tests/inductor/test_inductor_ops.py).

Prior error: InductorError: LoweringException: AssertionError in aten.cat.default lowering when one input was an empty 1D tensor (size [0], stride [1]) and the other a 4D fp16 tensor; failures were deterministic across May 24-27.

Assessment: Resolved persistent regression. Fix landed between 2026-05-27 and 2026-05-28 03:16 UTC. **No flakiness pattern.**

Recommended action: Confirm tomorrow's nightly stays green. Identify the fix commit if user wants to acknowledge the contributor.

Version Alignment

Check	Expected	Actual	Status
PyTorch	torch~=2.11.0 (pyproject.toml)	2.11.0+cpu	MATCH
transformers (post-build)	>=5.0 (hf_adapters)	4.57.6 after uv sync; bumped to 5.9.0 for E2E	DRIFT (POST-INSTALL)

Repo Drift (last 7 days)

Repo	Commits	API-touching	Risk
deeptools	82	10+ (DCC, BaseOpt, RcuOpt allocator, DataOp burst, senarch, DSM internals)	Medium — Heavy churn, but build still clean. Top: cdec8614d0 external/DDB cstdint, cb6cd81ce0 senulator reg-write delay, 18e61c1146 DCC nop reduction
flex	11	7 (FlexAllocator API, profiler API consolidation)	Medium — a233ca5 region_id→segment_id mapping, 5132737 profiler API consolidation, 8f06481 remove default directive from FlexAllocator::allocate
torch-spyre	63	—	—

Blocking PRs / Notable Open

PR	Title	Why Notable	Status
#2218	Upgrade to PT 2.12	Cross-repo PyTorch version bump	REVIEW_REQUIRED (7d open)
#2275	Replace dataFormat_ label in SDSC with fp32 for int32 identity	Touches SDSC/deeptools-adjacent label semantics	REVIEW_REQUIRED
#2107	Fix layernormscale output scale for fp32 LayerNorm	FP32 numeric correctness; fixes #1980	REVIEW_REQUIRED (13d, stalled)
#1943	Dualkineto integration	torch-profiler cross-component	CHANGES_REQUESTED (stalled)
#2003	Support PyTorch mark_dynamic API with max	PyTorch API compat / dynamic shapes	REVIEW_REQUIRED (17d, stalled)
#1440 / #1444	cpu_compile flags fix / linters.yaml typo	APPROVED, ready to merge — easy wins	APPROVED (50d, stalled)

Project Velocity (last 7 days)

Metric	Count
PRs merged	30+ (query hit 30-PR limit)
Open PRs (non-draft)	~58

Stalled PRs (no activity >7d)	30+
Top stalled by age	#1299 (dependabot, 03-31), #1440 (APPROVED, 04-08), #1444 (APPROVED, 04-08), #1576/#1577 (dependabot, 04-14)

Open Bugs

Metric	Value
Total open bugs	30
New (last 24h)	0
Filed (last 7d)	4 — #2247, #2246, #2222, #2207
Closed (last 7d)	13 — strong fix velocity
Net trend (7d)	-9 (closed 3x faster than filed)
Critical/priority labels	None set (no priority labels in use)

Recent Filings (last 7 days)

Issue	Title	Area
#2247	propagate_hints positional matching	Inductor
#2246	propagate_hints thread safety	Inductor
#2222	logical_not bool decomp	Op coverage
#2207	Mistral-Small unexpected stick expression	Model enablement

Smoke Test (post-build)

Stage	Result	Detail
Import torch_spyre	PASS	Spyre available=True, device_count=1
Eager add (fp16, 4×4)	PASS	shape=(4,4)
Compiled add (fp16, 128×128)	PASS	torch.compile() default Inductor backend on Spyre
E2E Qwen3 0.6B (5 tokens)	PASS	Load: 3.9s, Generate: 80.3s, Output: " Paris. The capital of"

Cross-Repo API Mismatches

None detected. All three components built clean against their respective HEADs (deeptools master, flex main, torch-spyre main). FlexAllocator API additions in flex landed without breaking torch-spyre callers.

